

Sentinel AI : AI-Driven Surveillance

Project Team

Abdullah Shahid	21I-0326
Ibrahim Abid	21I-0298
Sarib Ali	21I-0283

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Supervised by

Mr. Ahmad Raza

Co-Supervised by

Ms. Ayesha Marriyam



Department of Data Science and Artificial Intelligence

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

An AI system that can be deployed as a Software as a Service (SAAS) product on an already in-use CCTV camera network to provide the users with smart notifications along with an AI-enabled caller to contact emergency services or contacts on their behalf taking a step towards cost-efficient security. [1].

1.1 Problem Statement

Traditional CCTV systems provide only basic surveillance and lack advanced features such as intrusion detection. Upgrading to smart camera networks requires costly replacements, which is often not feasible for many users. This creates a gap in the market for affordable and scalable solutions that can enhance existing CCTV infrastructure without the need for full system overhauls.

1.2 Scope

This project aims to develop an AI-driven intruder detection system "Sentinel AI" that enhances existing CCTV infrastructure without costly hardware upgrades. Integrating advanced facial recognition capabilities to identify individuals from various angles and with different hairstyles ensures flexibility in real-world conditions, including a liveness detection model to prevent spoofing attempts using photos or videos. The project aims to categorize individuals into two groups: household members, who can move freely at any time and frequent visitors, who will trigger notifications if they arrive at unusual times. Real-time alerts will be generated for unfamiliar individuals or security breaches, immediately notifying the owner or even contacting emergency services through an AI-driven response system. Additionally, the solution will recognize faces even if partially

covered by masks, mufflers, or caps.

1.3 Modules

1.3.1 Face Recognition

1. Liveliness Detection
2. Face Detection from different angles
3. Face Detection with different hairstyles
4. Face Detection when the face is partially covered
5. Face addition of known personnel with permission of users
6. Intruder Detection

1.3.2 AI Caller

1. Emergency Notifications
2. Automatic Calls to Owner/Authorities

1.3.3 Web / App Development

1. Client Web App
2. Client Mobile App
3. Admin Web App

1.4 Tools and Technologies

- **Programming Languages**

- Python

- **Libraries Used**

- OpenCV

- Scikit-Learn
 - Pytorch
 - TensorFlow
- **Database/Framework**
 - MongoDB
 - Flask
 - React.JS
- **UI**
 - Flutter
- **Extras**
 - GitHub

1.5 User Classes and Characteristics

This section identifies the various user classes that will utilize the Sentinel AI system and describes their pertinent characteristics.

User Class	Description
Homeowner	The Homeowner is the primary user of the Sentinel AI system, responsible for setting up and managing the security features. They will interact with the system to configure settings, view alerts, and manage user permissions.
Visitors	Visitors are frequent guests of the household, such as friends or family members. With the Homeowner's consent, the system will add Visitors to the database for easier recognition, improving convenience while ensuring security.
Emergency Services	Emergency Services are external responders who receive alerts from the Sentinel AI system regarding potential security breaches. They will rely on accurate and timely notifications from the system to respond to incidents.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Event-Response Table

Below is an event-response table highlighting how the system reacts to various events:

Event	System Response
Face detected from different angles	Identify the individual using facial recognition and check if they match household members or frequent visitors. If unidentified, mark as a potential intruder.
Intruder detection	Trigger real-time notifications to the owner and initiate the AI-enabled caller to contact emergency services if required.
Face partially covered (mask, muffler, etc.)	Use partial facial recognition algorithms to attempt identification and categorize as household member, frequent visitor, or intruder.
Liveliness detection failure	Trigger an alert that a potential spoofing attempt (e.g., photo or video) was detected and log the attempt.
Emergency situation detected	Automatically initiate a call to emergency services with relevant details of the security breach.
Visitor added by owner	Add facial data for future detection with necessary permissions and update system database with user metadata.

Table 2.1: Event-Response Table for Sentinel AI System

2.2 Functional Requirements

The functional requirements describe what the system will do, grouped into three main modules: Face Recognition, AI Caller, and Web/App Development.

2.2.1 Face Recognition Module

1. The system shall detect faces from various angles, including front, left, and right views.
2. The system shall perform facial recognition even if the individual has changed their hairstyle or is partially covering their face (e.g., with a mask/muffler).
3. The system shall include liveness detection to ensure that the face is real and not a photo.
4. The system shall allow the owner to add known personnel's faces to the database with the appropriate permissions.
5. The system shall trigger an intruder alert for unfamiliar faces trying to enter the facility and log the event for further review.

2.2.2 AI Caller Module

1. The system shall automatically notify the owner via notifications when an intruder is detected.
2. The system shall automatically contact emergency services or designated contacts in case of a confirmed intruder or emergency situation.

2.2.3 Web/App Development Module

1. The system shall provide a client web app for the owner to monitor real-time video feeds, and review past events.
2. The system shall provide a client mobile app with similar functionality, enabling notifications and alerts on-the-go.

2.3 Non-Functional Requirements

This section specifies non-functional requirements for the Sentinel AI system. These quality requirements are specific, quantitative, and verifiable.

2.3.1 Reliability

The Sentinel AI system shall maintain a high level of reliability, with a Mean Time Between Failures (MTBF) of no less than 1000 hours. A failure is defined as any event that prevents the system from accurately detecting and notifying about intruders.

2.3.2 Usability

Usability requirements focus on providing an intuitive and efficient user experience. The system shall allow users to:

- Retrieve previous incident alerts with a single interaction.
- Access facial recognition settings and notifications without extensive training.

2.3.3 Performance

The Sentinel AI system shall meet the following performance requirements:

- 95% intruder alerts shall be processed within 2 seconds of detection.
- The system shall support a minimum of 100 concurrent users without degradation in performance.

2.3.4 Security

The system shall implement robust security measures to protect user data and system integrity. The security requirements include:

- The effort required to compromise the system shall be sufficiently high to deter unauthorized access.
- The system shall regularly audit user access and alert administrators of suspicious activity.

2.4 Domain Model

Create a representation of the domain model for your project.

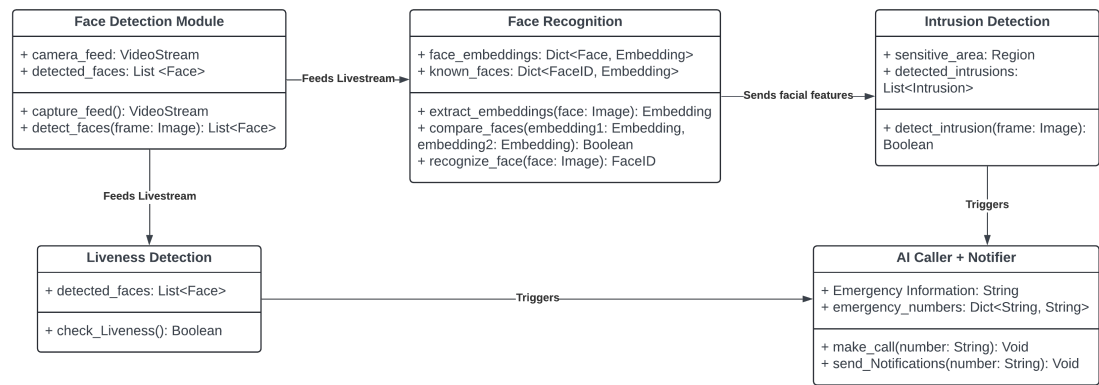


Figure 2.1: UML Model for Sentinel AI System

Chapter 3

System Overview

Sentinel AI is an AI-driven intruder detection system designed to enhance existing CCTV infrastructure without the need for expensive hardware upgrades. The system integrates facial recognition, liveness detection, and categorizes individuals into known and unknown categories to trigger real-time alerts in case of security breaches or intrusions. It can also notify emergency services and the property owner via AI-driven response systems. Sentinel AI is able to recognize faces even with masks, mufflers, or hats, and offers an efficient and scalable solution for residential and commercial security.

3.1 Architectural Design

The system architecture is modular, consisting of multiple subsystems that work together to achieve the complete functionality of Sentinel AI. Each subsystem has clearly defined responsibilities, and the system is decomposed into the following core modules:

- **Face Detection Module:** Responsible for capturing live video feeds and detecting faces in real time from multiple angles (front, left, right).
- **Liveness Detection Module:** Verifies whether the detected faces are real or fake, ensuring that spoofing attempts are caught.
- **Face Recognition Module:** Compares the detected face embeddings with known faces and identifies individuals based on stored embeddings.
- **Intrusion Detection Module:** Monitors sensitive areas within the camera feed for any unauthorized access or intrusions.
- **AI Caller & Notifier Module:** Alerts the property owner and emergency contacts by making automated calls and sending notifications in case of intrusion detection.

The modules interact as follows: The *Face Detection Module* continuously captures the camera feed and detects faces, passing them to the *Liveness Detection Module*. If a live face is detected, it is then sent to the *Face Recognition Module* for identification. If an unrecognized or unauthorized person is detected in a sensitive area, the *Intrusion Detection Module* triggers the *AI Caller & Notifier Module* to alert the relevant personnel.

The architectural style follows a layered approach, where each module operates independently but collaborates via well-defined interfaces. This modular approach allows flexibility and easy maintenance.

Below is a simplified box and line diagram showing the system's major subsystems and their connections:

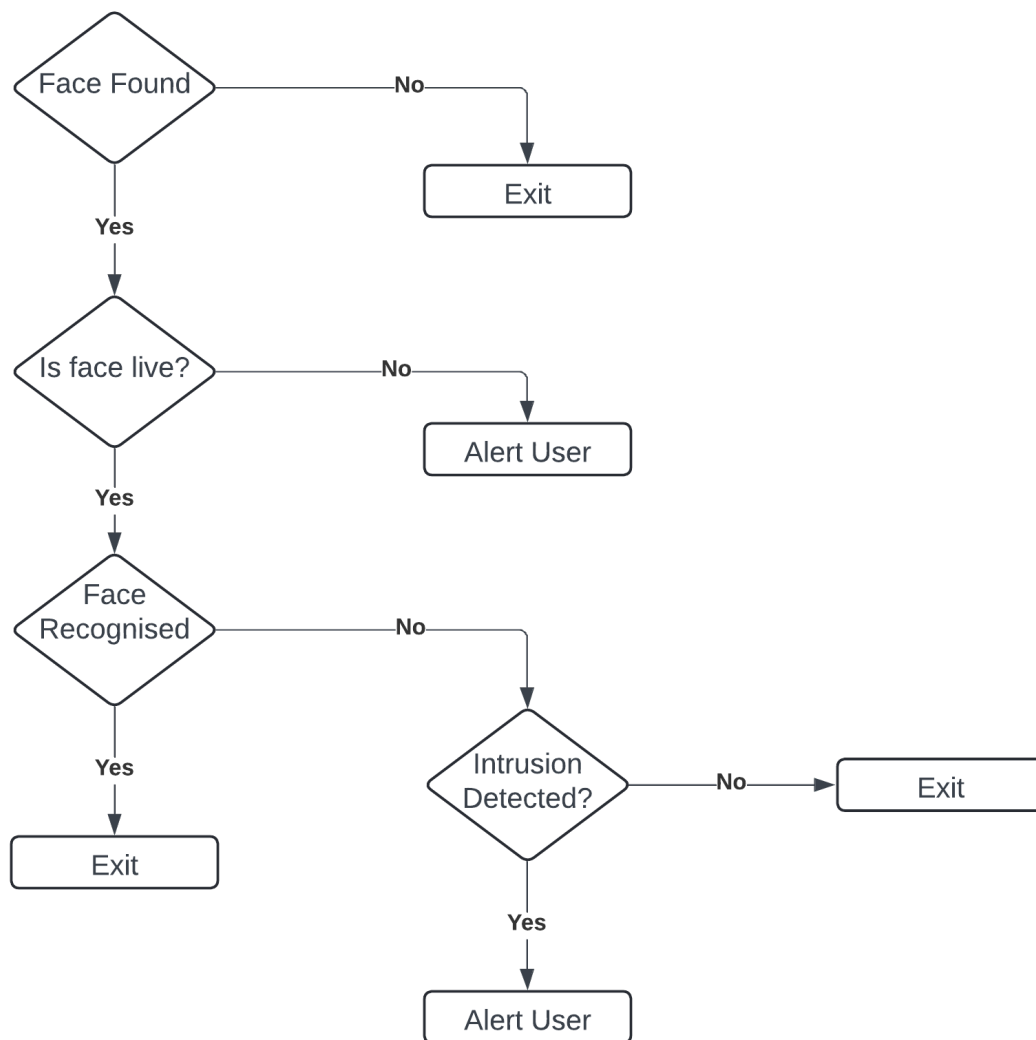


Figure 3.1: Box and Line Diagram of Sentinel AI's System Architecture

3.2 Design Models

The design of Sentinel AI follows an object-oriented approach, and the following models were created to represent different aspects of the system.

Activity Diagram:

The activity diagram shows the flow of activities within the system, starting from the camera feed capture to detecting an intrusion and sending out notifications. It represents how different modules interact to achieve system functionality.

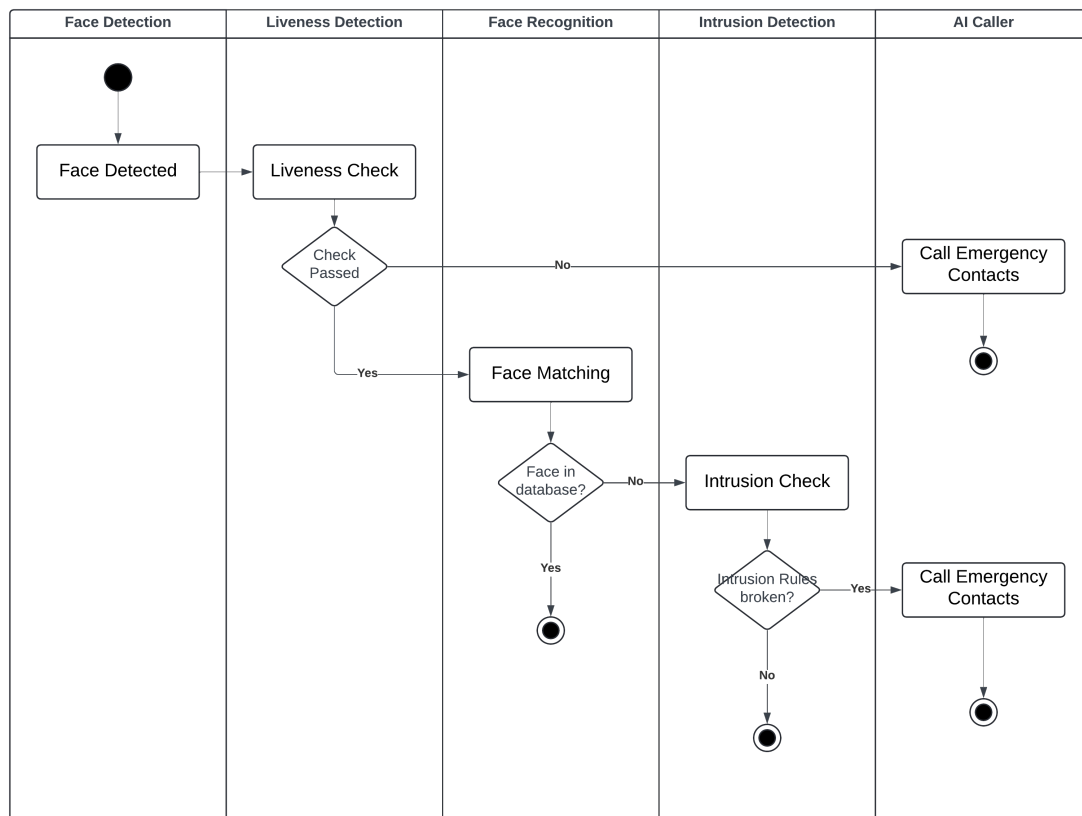


Figure 3.2: Activity Diagram of Sentinel AI's System Architecture

3.3 Data Design

Sentinel AI leverages a structured approach to handle and process data. The system stores data in two main components:

- **Face Embeddings Database:** This database stores the facial embeddings of known individuals, which are used for face recognition. The embeddings are extracted using deep learning techniques and stored as numerical vectors.

- **Event Logs and Intrusion Records:** Stores all detected intrusions, their timestamps, and details regarding the individuals involved. This is used for system audits and to provide a historical overview of intrusions.

The system uses MongoDB to store structured data (face embeddings, detected faces, and logs) in a highly scalable manner. The database is optimized for fast retrieval of face embeddings during the recognition process.

Additionally, emergency contact information is stored securely to ensure timely notification in case of security breaches.

Bibliography

- [1] Web Advisor and Chirag Aggarwal. Real-time anomaly detection in cctv surveillance. *Kaggle Datasets*, 2024.